

Binary search

`int[] a = new int[1 << 20]`

means

Here is left shifting binary 1 by 20 position.

$$1 \ll 20 = 2^{20} \\ = 1048576$$

a.length $\nearrow$

initially calls = 0

array becomes [0, 1, 2, 3, ..., 1048575]

static int binarysearch (---)

calls ++

-1
key =

if

lo = 0

lo > hi

hi = 1048575

0 > 1048575 X

-1 does not exist

not executed

$$\begin{aligned} \text{mid} &= 0 + (1048575 - 0) / 2 \\ &= 0 + 1048575 / 2 \\ &= 524287 \end{aligned}$$

compare with key

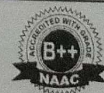
$a[524287] = 524287 \neq -1$ X not found mid

$-1 < 524287$ ✓

$\therefore$ lo = 0 hi = 524286



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at final search

mid=0

a[0]=0

~~0 = -1~~ ✗

not return mid

-1 < 0 ✓

hi = -1

it return -1

~~0~~

Fibonacci

for (int n=10;

n <= 30 10 <= 30 ✓

n += 5 n = 10 + 5 = 15

initially n = 10

static long fib()

calls++ - 0

n < 2

10 < 2 ✗

not executed

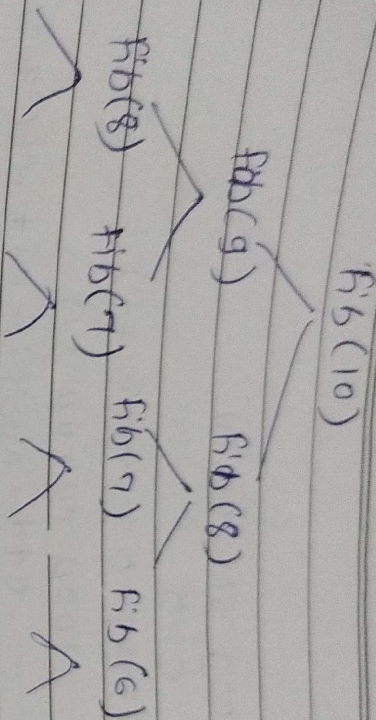
return

fib(n-1) + fib(n-2);

fib(10-1) + fib(10-2)

fib(9) + fib(8)





It calls up 177 $\text{fib}(10) = 55$

Similar $n = 10 + 5$
 $n = 15$

~~fib(15)~~

$\text{fib}(15) = 610$ calls = 1973

So they grow exponentially
time complexity is $O(n^2)$